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Complete

① $\ln 1 = \overset{0}{\cdots}$

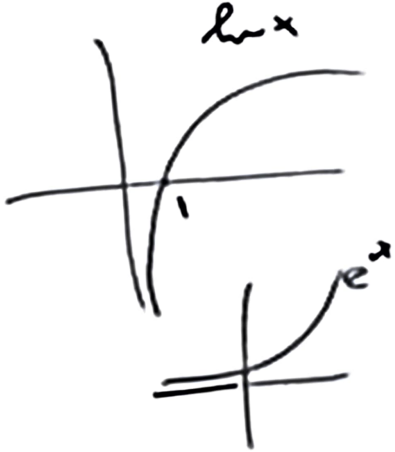
② $\ln \infty = \overset{\infty}{\cdots}$

③ $\ln 0 = \overset{-\infty}{\cdots}$

④ $e^{-\infty} = \overset{0}{\cdots}$

⑤ $\sinh x + \cosh x = \cdots$

⑥ $\sinh x - \cosh x = \cdots$



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⑤ $\sinh x + \cosh x = \dfrac{e^x - \cancel{e^{-x}}}{2} + \dfrac{e^x + \cancel{e^{-x}}}{2}$

$= e^x$

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(6) $\sinh x - \cosh x = \cdots$

(5) $\sinh x + \cosh x = \dfrac{e^{x} - \cancel{e^{-x}}}{2} + \dfrac{e^{x} + \cancel{e^{-x}}}{2}$

$= \dfrac{e^{x}}{2} + \dfrac{e^{x}}{2} = e^{x}$

(6) $\sinh x - \cosh x = \dfrac{e^{x} - e^{-x}}{2} - \left(\dfrac{e^{x} + e^{-x}}{2}\right)$

$= \frac{1}{2}\left[\cancel{e^{x}} - e^{-x} - \cancel{e^{x}} - e^{-x}\right]$

$\frac{1}{2}\left(-2e^{-x}\right) = -e^{-x}$

(7) $\sinh \ln x = \cdots$

$\dfrac{e^{\ln x} - e^{-\ln x}}{2} = \dfrac{x - \frac{1}{x}}{2} = \frac{1}{2}\left(x - \frac{1}{x}\right)$

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Inverse trig. fns

$$\sin^{-1}x \neq \frac{1}{\sin x}$$

$$\sin^{-1}\left(\frac{\pi}{2}\right) = \frac{\pi}{2} = \theta$$

$$\Rightarrow \sin\theta = \frac{1}{2} = \frac{\pi}{6}$$

$$\sin^{-1}x = y \Longrightarrow x = \sin y$$

$$-1 \le x \le 1 \qquad -\frac{\pi}{2} \le y \le \frac{\pi}{2}$$

$$\sin^{-1}(\sin x) = x \qquad \forall\ -\frac{\pi}{2} \le x \le \frac{\pi}{2}$$

$$\sin(\sin^{-1}x) = x \qquad \forall\ -1 \le x \le 1$$

$$\sin^{-1}\left(\sin\frac{\pi}{4}\right) = \frac{\pi}{4} \qquad -\frac{\pi}{2} < \frac{\pi}{4} < \frac{\pi}{2}$$

$$\sin^{-1}(\sin\pi) = \pi$$

$$\sin(\sin^{-1}\pi) = \pi \times$$

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* Inverse trig. fns

$$\sin^{-1}x \neq \frac{1}{\sin x}$$

$$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = ?? = \theta$$

$$\Rightarrow \sin\theta = \frac{1}{\sqrt{2}} = \frac{\pi}{4}$$

$$\sin^{-1}x = y \Longrightarrow x = \sin y$$

$$-1 \le x \le 1 \qquad -\frac{\pi}{2} \le y \le \frac{\pi}{2}$$

$$\sin^{-1}(\sin x) = x \quad \checkmark \; -\frac{\pi}{2} \le x \le \frac{\pi}{2}$$

$$\sin(\sin^{-1}x) = x \quad \checkmark \; -1 \le x \le 1$$

$\sin^{-1} x = y \implies x = \sin y$

$-1 \le x \le 1 \qquad -\frac{\pi}{2} \le y \le \frac{\pi}{2}$

$\sin^{-1}(\sin x) = x \qquad \forall\ -\frac{\pi}{2} \le x \le \frac{\pi}{2}$

$\sin(\sin^{-1} x) = x \qquad \forall\ -1 \le x \le 1$

$\sin^{-1}(\sin \frac{\pi}{4}) = \frac{\pi}{4} \qquad -\frac{\pi}{2} < \frac{\pi}{4} < \frac{\pi}{2}$

$\sin^{-1}(\sin 3) = \cancel{3}$

$\sin(\sin^{-1} 2) = 2 \times$

$\sin(\sin^{-1} \frac{1}{2}) = \frac{1}{2}$

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(2) $y = \cos^{-1} x \Longrightarrow x = \cos y$

$-1 \le x \le 1 \Longrightarrow 0 \le y \le \pi$

$\cos \cos^{-1} x = x \quad \forall \quad -1 \le x \le 1$

$\cos^{-1}(\cos x) = x \quad \forall \quad 0 \le x \le \pi$

$\cos^{-1} x \neq (\cos x)^{-1} = \frac{1}{\cos x} = \sec x$

(3) $y = \tan^{-1} x \Longrightarrow y = \tan y$

$-\infty < x < \infty \qquad -\frac{\pi}{2} < y < \frac{\pi}{2}$

① $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$.

proof

let $\sin^{-1}x = \alpha \Longrightarrow x = \sin\alpha$ ①

$\cos^{-1}x = \beta \Longrightarrow x = \cos\beta$ ②

from ① and ②

$\sin\alpha = \cos\beta$

$\therefore \alpha + \beta = \frac{\pi}{2}$

$\boxed{\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}}$

$$\sin(2\sin^{-1}x) = 2x\sqrt{1-x^2}$$

Sol

let $\sin^{-1}x = \alpha \Rightarrow \boxed{x = \sin\alpha}$ (*)

LHS

$\sin(2\sin^{-1}x) = \sin(2\alpha) = 2\sin\alpha\cos\alpha$ (1)

from * and **

$\Rightarrow$

$\because x = \sin\alpha$

$\Rightarrow \cos\alpha = \frac{\sqrt{1-}}{1}$ (**)

$\sin(2\sin^{-1}x) = 2\sin\alpha\cos\alpha$

$= 2x\sqrt{1-x^2} = R.H.S$

Prove

$$\cos(2\tan^{-1}x) = \frac{1-x^2}{1+x^2}.$$

let $\tan^{-1}x = \alpha \Rightarrow \boxed{x = \tan\alpha}$

L.H.S

$$\cos(2\tan^{-1}x) = \cos(2\alpha) = \cos^2\alpha - \sin^2\alpha$$

$$= \left(\frac{1}{\sqrt{1+x^2}}\right)^2 - \left(\frac{x}{\sqrt{1+x^2}}\right)^2$$

$\because \frac{x}{1} = \tan\alpha$

$$= \frac{1}{1+x^2} - \frac{x^2}{1+x^2} = \frac{1-x^2}{1+x^2} = \text{R.H.S.}$$

Prove that

$$\sinh^{-1}x = \ln(x+\sqrt{x^2+1})$$

Sol

let $\sinh^{-1}x = y \Rightarrow x = \sinh y$

$\therefore x = \dfrac{e^y - e^{-y}}{2} \Rightarrow 2x = e^y - e^{-y}$

$e^y \times$

$2xe^y = e^{2y} - 1$

$\Rightarrow e^{2y} - 2xe^y - 1 = 0$

$(e^y)^2 - 2x(e^y) - 1 = 0$

$\therefore e^y = \dfrac{+2x \pm \sqrt{(2x)^2 - 4(-1)}}{2(1)}$

$e^y = \dfrac{2x \pm \sqrt{4x^2+4}}{2} = \dfrac{\cancel{2}(x \pm \cancel{2}\sqrt{x^2+1})}{\cancel{2}}$

$= x \pm \sqrt{x^2+1}$

$\Rightarrow$

$ax^2 + bx + c = 0$

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$(e^y)^2 - 2x(e^y) - 1 = 0$

$a = 1, \quad b = -2x, \quad c = -1$

$$(e^{y})^2 - 2x(e^{y}) - 1 = 0$$

$$e^{y} = \frac{+2x \pm \sqrt{(2x)^2 - 4(-1)}}{2(1)}$$

$$(e^{y})^2 - 2x(e^{y}) - 1 = 0$$

$$a = 1,\quad b = -2x,\quad c = -1$$

$$e^{y} = \frac{2x \pm \sqrt{4x^2+4}}{2} = \frac{\cancel{2}(x \pm \cancel{2}\sqrt{x^2+1})}{\cancel{2}}$$

$$= \frac{2x \pm 2\sqrt{x^2+1}}{2}$$

$$\Rightarrow e^{y} = x \pm \sqrt{x^2+1}$$

لو فاتح الطرفين

$$\Rightarrow y = \ln\left(x \pm \sqrt{x^2+1}\right)$$

$$\Rightarrow \sinh^{-1} x = \ln\left(x \pm \sqrt{x^2+1}\right)$$

$$\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$$ or $$\sinh^{-1} x = \ln\left(x - \sqrt{x^2+1}\right)$$

مرفوض

$\sqrt{x^2+1} > x$ دائماً

مرفوض لأن

$\Rightarrow \quad \sqrt{x^2+1}$

لو فاتح الطريق

$$\Rightarrow \quad y = \ln\left(x \pm \sqrt{x^2+1}\right)$$

$$\Rightarrow \quad \sinh^{-1} x = \ln\left(x \pm \sqrt{x^2+1}\right)$$

$\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$ or $\sinh^{-1} x = \ln\left(x - \sqrt{x^2+1}\right)$

مرفوض

$\sqrt{x^2+1} > x$ دائماً

$\therefore \quad \sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$

Prove

① $\operatorname{csch}^{-1} x = \ln\left(x \pm \sqrt{x^2-1}\right)$

② $\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

② let $\tanh^{-1} x = y$

$\Rightarrow x = \tanh y = \frac{\sinh y}{\cosh y} = \frac{e^{y} - e^{-y}}{e^{y} + e^{-y}}$

$\Rightarrow x = \boxed{\frac{e^{y} - e^{-y}}{e^{y} + e^{-y}}}$ ÷ بالبسط و المقام

$\therefore x = \frac{e^{2y} - 1}{e^{2y} + 1}$

$\therefore e^{2y} - 1 = x e^{2y} + x$

$e^{2y} - x e^{2y} = x + 1$

$e^{2y}(1-x) = x+1$

$\Rightarrow e^{2y} = \frac{1+x}{1-x}$

$$\Rightarrow \quad x = \boxed{\frac{e^{2y}-1}{e^{2y}+1}}$$

بالضرب في 2

$$\therefore \quad x = \frac{e^{2y}-1}{e^{2y}+1}$$

$$\therefore \quad e^{2y}-1 = x e^{2y} + x$$

$$e^{2y} - x e^{2y} = x+1$$

$$e^{2y}(1-x) = x+1$$

$$\Rightarrow e^{2y} = \frac{1+x}{1-x}$$

$$e^{2y}(1-x) = x+1 \Rightarrow e^{2y} = \frac{1+x}{1-x}$$

إذن ln الطرفين

$$\ln(e^{2y}) = \ln\left(\frac{1+x}{1-x}\right)$$

$$2y = \ln\left(\frac{1+x}{1-x}\right) \Rightarrow y = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$= \tanh^{-1}x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$x \to 0^+$ $\quad$ $x \to 0^-$

for $f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$

$\lim_{x \to 5} f(x) = 1$

$\therefore \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} 1 = 1$

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} 0 = 0$

$\lim_{x \to 0} f(x)$ غير موجودة

$\lim_{x \to 0^+} f(x) \neq \lim_{x \to 0^-} f(x)$

$\lim_{x \to 2} \frac{x^2-4}{x-2} = \frac{0}{0}$

$= \lim_{x \to 2} \frac{(x+2)(x-2)}{(x-2)}$

$= \lim_{x \to 2} (x+2)$

$= \boxed{4}$

Basic properties of limits

* If $\lim_{x \to a} f(x) = L$, $\lim_{x \to a} g(x) = M$

then

$$\lim_{x \to a} (f(x) \pm g(x)) = \lim_{x \to a} f \pm \lim_{x \to a} g$$

$$= L \pm M$$

$$\lim_{x \to 1} x^2 - x = 2$$

$$\lim_{x \to 0} \sqrt{x+1} \pm \sin x$$

$$\lim_{x \to a} (f(x) g(x))$$

$$= \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$$

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$$\lim_{x\to a}\left(f(x)g(x)\right)$$

$$= \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$$

$$* \quad \lim_{x\to a} \ln f(x) = \ln \lim_{x\to a} f(x)$$

$$* \quad \lim_{x\to a}\left(f(x)\right)^n = \left(\lim_{x\to a} f(x)\right)^n$$

$$\lim \left(1+\frac{x}{n}\right)^n$$

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$$

$$= \lim_{n\to 0}\left(1+nx\right)^{\frac{1}{n}}$$

$\lim_{x \to 0} x$ 282 554 $\lim_{x \to a} = g(x)$

$$* \quad \lim_{x \to a} \ln f(x) = \ln \lim_{x \to a} f(x)$$

$$* \quad \lim_{x \to a} (f(x))^n = \left(\lim_{x \to a} f(x)\right)^n$$

$$e^x = \lim_{n \to \infty} \left(1 + \frac{x}{n}\right)^n$$

$$= \lim_{n \to 0} (1 + nx)^{\frac{1}{n}}$$

$\underline{\underline{\text{ex}}}$ find $\lim_{n \to \infty} \left(1 + \frac{3}{n}\right)^{2n}$

$$= \left(\lim_{n \to \infty} \left(1 + \frac{3}{n}\right)^n\right)^2 = (e^3)^2 = e^6$$

Evaluate

$$\lim_{n\to\infty}\left(1-\frac{3}{4n}\right)^{n-1}$$

$$x^3 \cdot x^n = x^{3+n}$$

$$= \lim_{n\to\infty}\left(\left(1-\frac{3}{4n}\right)^{n}\cdot\left(1-\frac{3}{4n}\right)^{-1}\right)$$

$$= \lim_{n\to\infty}\left(1-\frac{3}{4n}\right)^{n}\cdot\lim_{n\to\infty}\left(1-\frac{3}{4n}\right)^{-1}$$

$$1$$

$$\lim_{n\to\infty}\left(1-\frac{3}{n}\right)^{n} = e^{-3/4}$$

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^{n}$$

$$\lim_{n\to\infty}\left(1-\frac{3}{n}\right)^{n} = e^{-3/4}$$

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Evaluate

$$\lim_{x\to\infty} x\left[\ln(x+5)-\ln x\right]$$

$$\ln a - \ln b = \ln\frac{a}{b}$$

$$a\ln b = \ln b^{a}$$

$$= \lim_{x\to\infty} x\left[\ln\frac{x+5}{x}\right]$$

$$= \lim_{x\to\infty} \ln\left(\frac{x+5}{x}\right)^{x}$$

$$= \lim_{x\to\infty} \ln\left(1+\frac{5}{x}\right)^{x}$$

$$= \ln \lim_{x\to\infty}\left(1+\frac{5}{x}\right)^{x} = \ln e^{5} = 5$$

Evaluate

$$\lim_{n\to\infty}\left(\frac{n-3}{n+4}\right)^{2n-7}$$

$$= \lim_{n\to\infty}\left(\frac{n-3}{n+4}\right)^{2n} \cdot \lim_{n\to\infty}\left(\frac{n-3}{n+4}\right)^{-7}$$

$$= \left(\lim_{n\to\infty}\left(\frac{1-\frac{3}{n}}{1+\frac{4}{n}}\right)^{n}\right)^{2}$$